Loops and Iterations – MCQ Assignment

Instructions: Choose the correct answer from the four options given for each question.

1. What is the main purpose of a loop in programming?

A) To store data
B) To repeat a block of code
C) To create a variable
D) To stop a program

2. Which loop is commonly used when we know how many times a task must be repeated?

A) for loop
B) if statement
C) switch statement
D) break statement

3. Which loop checks its condition before running the loop body?

A) do-while loop
B) while loop
C) Both while and do-while loops
D) None of these

4. What is the minimum number of times a do-while loop runs?

A) Zero times
B) One time
C) Two times
D) It depends on the condition

5. What happens when a break statement runs inside a loop?

A) The loop starts again
B) The loop ends immediately
C) The program runs forever
D) The condition becomes true

6. What does the continue statement do inside a loop?

A) Stops the whole program
B) Skips the current iteration and moves to the next one
C) Ends the loop permanently
D) Restarts the program

7. What is a nested loop?

A) A loop inside another loop
B) A loop without a condition
C) A loop that runs only once
D) A loop that never ends

8. Which loop can be used to print a multiplication table?

A) for loop
B) if statement
C) break statement
D) return statement

9. What is the factorial of a positive integer?

A) The sum of numbers from 1 to that number
B) The product of positive integers from 1 to that number
C) The square of that number
D) The number multiplied by 2

10. Which of the following numbers is a prime number?

A) 9
B) 15
C) 17
D) 21

11. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i < 5; i++)
        cout << i << " ";
    return 0;
}
```

A) 0 1 2 3 4
B) 1 2 3 4
C) 1 2 3 4 5
D) 0 1 2 3

12. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    while (i <= 3) {
        cout << i << " ";
        i++;
    }
    return 0;
}
```

A) 1 2 3
B) 1 2 3 4
C) 0 1 2 3
D) 3 2 1

13. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 5; i++) {
        if (i == 3) break;
        cout << i << " ";
    }
    return 0;
}
```

A) 1 2
B) 1 2 3
C) 1 2 4 5
D) 1 2 3 4 5

14. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 5; i++) {
        if (i == 3) continue;
        cout << i << " ";
    }
    return 0;
}
```

A) 1 2 3 4 5
B) 1 2 4 5
C) 1 2
D) 3 4 5

15. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int total = 0;
    for (int i = 1; i < 5; i++) total += i;
    cout << total;
    return 0;
}
```

A) 4
B) 6
C) 10
D) 15

16. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int n = 5, fact = 1;
    for (int i = 1; i <= n; i++) fact *= i;
    cout << fact;
    return 0;
}
```

A) 15
B) 25
C) 120
D) 5

17. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int n = 1234, count = 0;
    while (n > 0) {
        count++;
        n /= 10;
    }
    cout << count;
    return 0;
}
```

A) 3
B) 4
C) 5
D) 10

18. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int n = 123, total = 0;
    while (n > 0) {
        total += n % 10;
        n /= 10;
    }
    cout << total;
    return 0;
}
```

A) 5
B) 6
C) 123
D) 321

19. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 3; i++) {
        for (int j = 1; j <= 2; j++)
            cout << "* ";
        cout << endl;
    }
    return 0;
}
```

A) * on three separate lines
B) * * on each of three lines
C) * * * on each of two lines
D) * * on one line

20. What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
int main() {
    int n = 121, rev = 0, temp = n;
    while (temp > 0) {
        int digit = temp % 10;
        rev = rev * 10 + digit;
        temp /= 10;
    }
    if (n == rev) cout << "Palindrome";
    else cout << "Not Palindrome";
    return 0;
}
```

A) 121

B) Not Palindrome

C) Palindrome

D) 12